

CATFISH FRY SIZE CLASSIFICATION

Using Convolutional Neural Network (CNN) and OpenMV Cam H7+

English Portfolio Report

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Executive Summary

This report presents an English portfolio version of a final thesis project titled "Towards Automatic Catfish Fry Grading: Catfish Fry Size Classification Using Convolutional Neural Network (CNN) with OpenMV Cam." The project explored the use of computer vision and deep learning to classify catfish fry into size categories automatically.

The research used CNN-based image classification with MobileNetV2 transfer learning, TensorFlow/Keras training, TensorFlow Lite conversion, and OpenMV Cam H7+ deployment. The strongest configuration was 128 x 128 blobs.rect, which achieved 89% accuracy, 87% precision, 83% recall, and 82% F1-score in real-time testing.

1. Problem Background

Catfish fry grading is an important process in catfish farming. Fry can grow at different rates even when they are of similar age. If fry of different sizes are mixed, smaller fry can be affected by cannibalism and buyers may receive fry that do not match the intended size category.

Traditional grading commonly uses manual sorting tools such as perforated containers or grading trays. However, manual grading can still produce inconsistent results when fry that do not match the target size pass through the sorting holes. This can create losses for both farmers and buyers.

The project addressed this problem by applying computer vision and CNN-based classification to support a more consistent and automated grading process.

2. Research Objective

The main objective was to apply deep learning with CNN to classify catfish fry sizes automatically using images captured through an OpenMV-based workflow.

- Apply CNN to the catfish fry grading problem.
- Classify catfish fry into predefined grade categories.
- Evaluate model performance through static image testing and real-time OpenMV testing.
- Identify the best image configuration for this grading task.

3. Scope and Size Classes

The research focused on catfish fry classification based on size categories. The fry used in the research were approximately 20 days old. The experimental setup used an aquarium and an OpenMV Cam H7+ camera.

Class	Size Range
Grade A	1 - 3 cm
Grade B	4 - 5 cm
Grade C	> 6 cm

4. Hardware and Software Setup

Area	Technology / Component
Camera	OpenMV Cam H7+
Camera position	Approximately 20 cm above the aquarium
Aquarium size	20 cm x 15 cm x 15 cm
Water depth	Approximately 0.5 cm

Model architecture	MobileNetV2 transfer learning
Training framework	TensorFlow / Keras
Deployment format	TensorFlow Lite (.tflite)
Deployment tool	OpenMV IDE

5. Methodology

The workflow combined dataset collection, image preprocessing, model training, model conversion, and embedded testing. The model was trained using TensorFlow/Keras, then converted into TensorFlow Lite format for OpenMV-based testing.

1. Capture catfish fry videos/images using the OpenMV camera setup.
2. Extract image frames and prepare dataset folders by grade category.
3. Preprocess images using different image sizes and blob formats.
4. Train the model using MobileNetV2 transfer learning.
5. Evaluate the model using classification metrics.
6. Convert the trained model into TensorFlow Lite format.
7. Test the model using static images and real-time OpenMV inference.

6. Dataset and Experimental Configurations

The research compared multiple image configurations to understand which preprocessing and image format provided stronger model performance.

Configuration	Description
96 x 96 blobs.fit	Image size 96 x 96 using blobs.fit
96 x 96 blobs.rect	Image size 96 x 96 using blobs.rect
128 x 128 blobs.fit	Image size 128 x 128 using blobs.fit
128 x 128 blobs.rect	Image size 128 x 128 using blobs.rect

Dataset Variant	Total Images	Notes
Fit	630	Earlier dataset configuration
Rectangle	1,740	Improved dataset configuration with stronger performance

7. Evaluation Metrics

The model was evaluated using confusion matrix representation and common classification metrics: accuracy, precision, recall, and F1-score. Static image testing evaluated the trained model on image samples, while real-time testing validated the OpenMV-based workflow.

8. Results

The best-performing configuration was 128 x 128 blobs.rect. It performed better than blobs.fit in both static image testing and real-time testing. The real-time result is the strongest portfolio highlight because it demonstrates that the model was tested in an embedded camera workflow, not only in offline static evaluation.

Test Type	Accuracy	Precision	Recall	F1-score
Static image testing - 128 x 128 blobs.rect	80%	70%	70%	68%

Real-time testing - 128 x 128 blobs.rect	89%	87%	83%	82%
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In real-time testing, the 128 x 128 blobs.rect configuration achieved average accuracy of 0.89, precision of 0.87, recall of 0.83, and F1-score of 0.82. This result shows that the model was able to identify grade differences with reasonably strong performance in the selected experimental environment.

9. Portfolio Value

This project is a strong portfolio item because it combines original research, applied machine learning, computer vision, embedded deployment, and formal academic reporting. Compared with typical course projects, this thesis demonstrates a complete technical workflow from problem formulation to model evaluation and deployment testing.

- Computer vision problem formulation in an agriculture/aquaculture context.
- CNN and transfer learning implementation using MobileNetV2.
- Image dataset preparation and preprocessing comparison.
- Model evaluation using accuracy, precision, recall, F1-score, and confusion matrix.
- TensorFlow Lite conversion and embedded testing through OpenMV Cam H7+.
- Formal thesis documentation and measurable experimental results.

10. Limitations and Future Improvements

The research was conducted within a controlled experimental setup. Future development can improve robustness by expanding the dataset, increasing variation in lighting and water conditions, testing different camera positions, and validating the model across more farms or aquarium environments.

- Increase dataset size and improve class balance.
- Capture more variation in lighting, water clarity, fish movement, and aquarium conditions.
- Improve real-time counting logic and object isolation.
- Test the model on unseen locations and real farm conditions.
- Explore newer lightweight model architectures for embedded deployment.

11. Closing

The project demonstrates that CNN-based computer vision can be applied to catfish fry grading using MobileNetV2 and OpenMV Cam H7+. The 128 x 128 blobs.rect configuration produced the strongest result and became the most suitable configuration for portfolio presentation.

As a portfolio artifact, this project supports professional positioning in computer vision, machine learning, Python, data preprocessing, embedded AI experimentation, and technical documentation.

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